241501006
ABHISHEK S

Exp No: 1	*Study of various Network commands used in Linux and Windows*

AIM:

To study and practise the various network commands used in Linux and Windows operating systems.

ALGORITHM / PROCEDURE:

1.Open the Terminal in Linux and the Command Prompt (cmd) in Windows.

2. Use hostname to see the name of the computer.

3.Use ifconfig (Linux) and ipconfig (Windows) to see the IP address, subnet mask and MAC address.

4. Use ping to check whether the remote system is reachable. Note the packet loss and the time taken.

5.Use traceroute (Linux) and tracert (Windows) to see the path taken by the packets.

6 Use nslookup and dig to find the IP address of a domain name.

. Use netstat to see the active connections and open ports.

7 Use arp -a to see the IP address to MAC address table.

9.Use route (Linux) and route print (Windows) to see the routing table.

8 10.Note down the output of each command.

.

Commands used:

Purpose	*Linux*	*Windows*
Host name	`hostname`	`hostname`
IP configuration	`ifconfig / ip addr`	`ipconfig /all`
Check connectivity	`ping -c 4 google.com`	`ping google.com`
Trace the route	`traceroute google.com`	`tracert google.com`
DNS lookup	`nslookup / dig`	`nslookup`
Active connections	`netstat -tuln`	`netstat -an`
ARP table	`arp -a`	`arp -a`
Routing table	`route -n`	`route print`
MAC address	`ip link`	`getmac`

DIAGRAM:

Fig 1: Network commands and the layers they work in

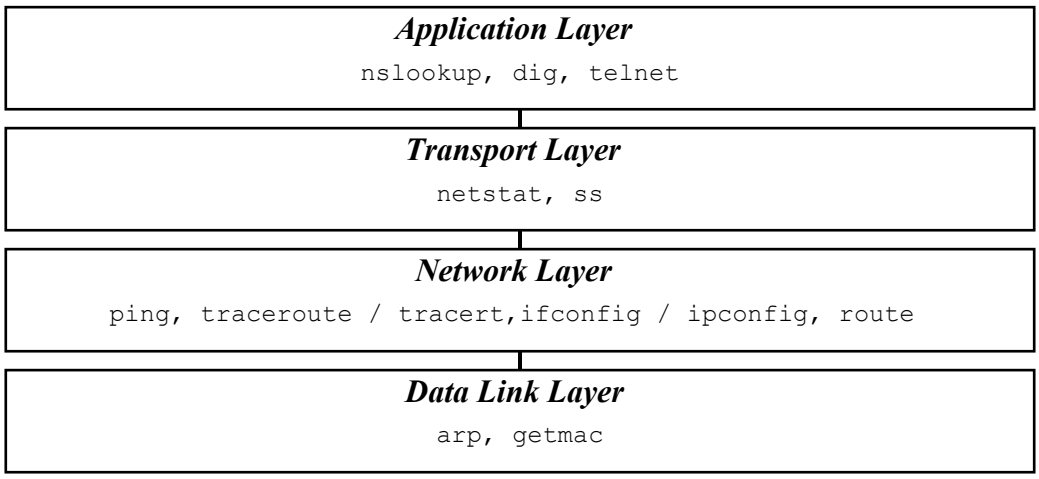
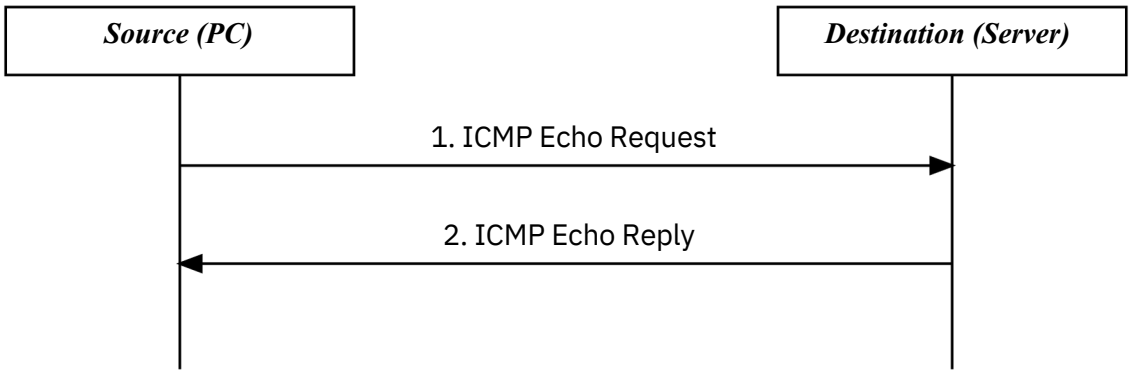


Fig 2: Working of ping command



OUTPUT:

Linux:

```
UusSeErR@UusSeErR-VirtualBox: ~

UusS    @UusSeErR    -VirtualBox:~$ hostname
eErR    -VirtualBox
UusS
```

```
UusSeErR@UusSeErR-VirtualBox: ~

UusSeErR@UusSeErR    -VirtualBox:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
        inet 192.168.1.10 netmask 255.255.255.0 broadcast 192.168.1.255
        inet6 fe80::a00:27ff:fe4e:66a1 prefixlen 64 scopeid 0x20<link>
        ether 08:00:27:4e:66:a1 txqueuelen 1000 (Ethernet)
        RX packets 2543 bytes 2154321 (2.1 MB)
        TX packets 1987 bytes 312456 (312.4 KB)

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
        inet 127.0.0.1 netmask 255.0.0.0
```

```
UusSeErR@UusSeErR-VirtualBox: ~

UusSeErR@UusSeErR-VirtualBox:~$ ping -c 4 google.com
PING google.com (142.250.195.78) 56(84) bytes of data.
64 bytes from maa03s45-in-f14.1e100.net (142.250.195.78): icmp_seq=1 ttl=117 time=14.2 ms
64 bytes from maa03s45-in-f14.1e100.net (142.250.195.78): icmp_seq=2 ttl=117 time=14.8 ms
64 bytes from maa03s45-in-f14.1e100.net (142.250.195.78): icmp_seq=3 ttl=117 time=13.9 ms
64 bytes from maa03s45-in-f14.1e100.net (142.250.195.78): icmp_seq=4 ttl=117 time=15.2 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 13.904/14.525/15.230/0.512 ms
```

```
UusSeErR@UusSeErR-VirtualBox: ~

UusSeErR@UusSeErR-VirtualBox:~$ traceroute google.com
traceroute to google.com (142.250.195.78), 30 hops max, 60 byte packets
 1 _gateway (192.168.1.1) 1.102 ms 1.045 ms 0.987 ms
 2 10.20.0.1 (10.20.0.1) 4.321 ms 4.287 ms 4.410 ms
 3 172.16.5.9 (172.16.5.9) 8.765 ms 8.801 ms 8.730 ms
 4 72.14.215.85 (72.14.215.85) 12.543 ms 12.498 ms 12.610 ms
 5 maa03s45-in-f14.1e100.net (142.250.195.78) 14.230 ms 14.187 ms 14.301 ms
```

```
UusSeErR@UusSeErR-VirtualBox: ~

UusSeErR@UusSeErR-VirtualBox:~$ nslookup google.com
Server:         192.168.1.1
Address:        192.168.1.1#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.195.78
```

```
UusSeErR@UusSeErR-VirtualBox: ~

UusSeErR@UusSeErR-VirtualBox:~$ dig google.com +short
142.250.195.78
```

```
UusSeErR@UusSeErR-VirtualBox: ~

UusSeErR@UusSeErR-VirtualBox:~$ netstat-tuln
Active Internet connections (onlyservers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN
tcp6       0      0 :::80                   :::*                    LISTEN
udp        0      0 0.0.0.0:68              0.0.0.0:*
```

```
UusSeErR@UusSeErR-VirtualBox: ~

UusSeErR@UusSeErR-VirtualBox:~$ arp -a
_gateway (192.168.1.1) at c4:e9:84:2a:5b:10 [ether] on enp0s3
? (192.168.1.12) at 3c:52:82:7d:11:9e [ether] on enp0s3
```

```
UusSeErR@UusSeErR-VirtualBox: ~

UusSeErR@UusSeErR-VirtualBox:~$ route -n
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
0.0.0.0         192.168.1.1     0.0.0.0         UG    100    0        0 enp0s3
192.168.1.0     0.0.0.0         255.255.255.0   U     100    0        0 enp0s3
```

Windows:

Command Prompt

```
C:\Users\UusSeErR>hostname
DESKTOP-UusSeErR
```

Command Prompt

```
C:\Users\UusSeErR>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : DESKTOP-UusSeErR
   Primary Dns Suffix . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No

Ethernet adapter Ethernet:

   Physical Address. . . . . . . . . : 00-1A-2B-3C-4D-5E
   DHCP Enabled. . . . . . . . . . . : Yes
   IPv4 Address. . . . . . . . . . . : 192.168.1.25(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.1.1
   DHCP Server . . . . . . . . . . . : 192.168.1.1
   DNS Servers . . . . . . . . . . . : 8.8.8.8
```

Command Prompt

```
C:\Users\UusSeErR>ping google.com

Pinging google.com [142.250.195.78] with 32 bytes of data:
Reply from 142.250.195.78: bytes=32 time=15ms TTL=117 Reply
from 142.250.195.78: bytes=32 time=14ms TTL=117 Reply from
142.250.195.78:  bytes=32  time=14ms  TTL=117  Reply  from
142.250.195.78: bytes=32 time=16ms TTL=117

Ping statistics for 142.250.195.78:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 14ms, Maximum = 16ms, Average = 14ms
```

Command Prompt

```
C:\Users\UusSeErR>tracert google.com

Tracing route to google.com [142.250.195.78]
over a maximum of 30 hops:

  1    <1ms     <1ms     <1ms  192.168.1.1
  2     4ms      4ms      4ms  10.20.0.1
  3     9ms      8ms      9ms  172.16.5.9
  4    13ms     12ms     12ms  72.14.215.85
  5    14ms     14ms     15ms  142.250.195.78

Trace complete.
```

Command Prompt

```
C:\Users\UusSeErR>nslookup google.com
Server: dns.google
Address: 8.8.8.8

Non-authoritative answer:
Name:    google.com
Address: 142.250.195.78
```

Command Prompt

```
C:\Users\UusSeErR>netstat -an

Active Connections

  Proto  Local      Address     Foreign    Address     State
  TCP    0.0.0.0:135            0.0.0.0:0              LISTENING
  TCP    0.0.0.0:445            0.0.0.0:0              LISTENING
  TCP    192.168.1.25:50234     142.250.195.78:443     ESTABLISHED
  UDP    0.0.0.0:5353           *:*
```

Command Prompt

```
C:\Users\UusSeErR>arp -a

Interface: 192.168.1.25 --- 0x8
 Internet Address       Physical  Address     Type
 192.168.1.1            c4-e9-84-2a-5b-10     dynamic
 192.168.1.255          ff-ff-ff-ff-ff-ff     static
```

Command Prompt

```
C:\Users\UusSeErR>route print

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface Metric
          0.0.0.0          0.0.0.0      192.168.1.1    192.168.1.25     25
        127.0.0.0        255.0.0.0         On-link         127.0.0.1    331
      192.168.1.0    255.255.255.0         On-link      192.168.1.25    281
```

Command Prompt

```
C:\Users\UusSeErR>getmac

Physical Address    Transport Name
=================== ==========================================================
00-1A-2B-3C-4D-5E   \Device\Tcpip_{8F3A21B0-5C6D-4E7F-9A1B-2C3D4E5F6A7B}
```

RESULT:

Thus the various network commands used in Linux and Windows were studied and executed successfully.